

## Problem Set 3 Exercise #08: Find Twice

**Reference:** Lecture 6 notes

**Learning objective:** One-dimensional array

**Estimated completion time:** 35 minutes

### Problem statement:

[CS1010E AY2010/11 Semester 2 Exam, Q16(a)]

Write a program **PS3\_Ex08\_FindTwice.java**.

Your program should contain a static method

```
int findTwice(int[] arr)
```

that determines if any integer is repeated in the array. If so, this method returns the least index where the repeated integer resides; if not, it returns -1.

For example, it will return 1 for the array {1, 3, 5, 5, 4} because 3 is the first repeated integer in the array and index of its first appearance is 1.

A tip is given at the end of this page.

### Sample run #1:

```
Enter the number of elements: 5
Enter 5 elements: 1 3 5 7 9
Array doesn't contain duplicated data
```

### Sample run #2:

```
Enter the number of elements: 5
Enter 5 elements: 1 3 5 5 3
3 is the first element that appears at least twice
```

### Useful tip:

Avoid duplicated comparisons. For example, if **arr[m]** has already been compared to **arr[n]**, then you shouldn't compare **arr[n]** to **arr[m]**.